



Code execution in peer review for your favourite conference or journal with CODECHECK

NFDI_BB meeting May 2026 (doi.org/10.25798/gmzb-cd87)

2026-05 | Daniel Nüst | orcid.org/0000-0002-0024-5046

Slides: doi.org/10.5281/zenodo.20084769

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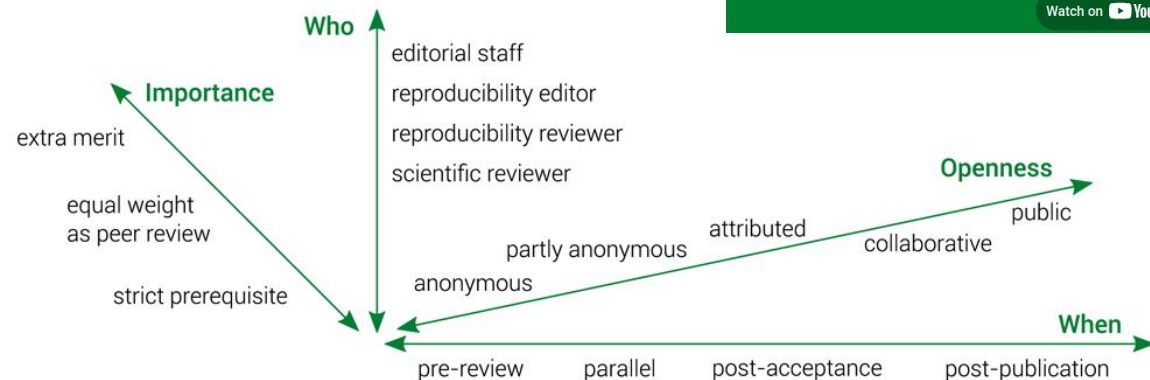
One re-execution of computational workflow by codechecker during scholarly communication, review & publication



Independent execution of computations underlying research articles.



1. Codecheckers record but don't investigate or fix.
2. Communication between humans is key.
3. Credit is given to codecheckers.
4. Workflows must be auditable.
5. Open by default and transitional by disposition.



100+ Certificates
<https://codecheck.org.uk/register/>

AGILE conference success story

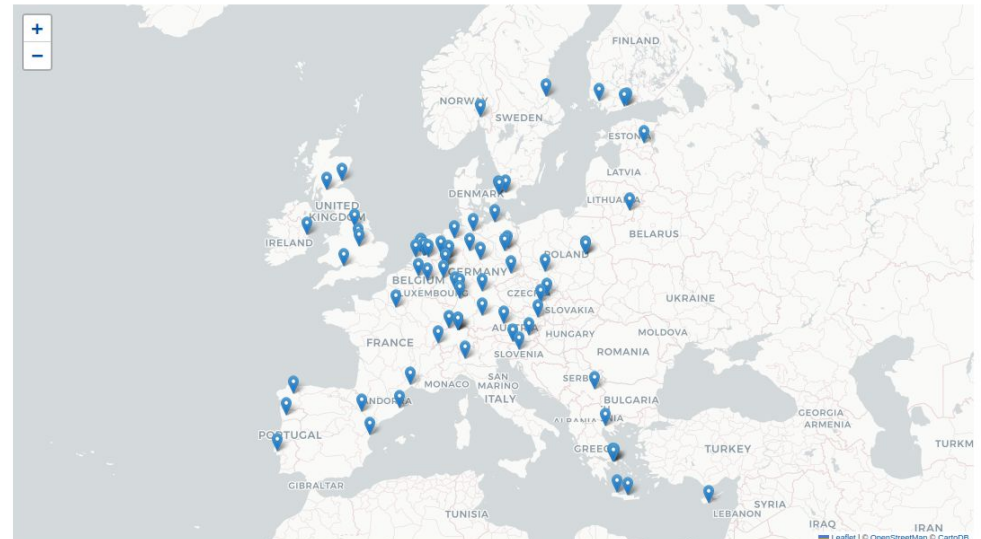
From zero to reproducible in
countable steps and observable time

Members

Search for: in AGILE memberbase [GO](#)

Name	City	Country
52 North	Muenster	Germany
Bell Edwards Geographic Data Institute, School of Geography & Sustainable Development, University of St Andrews	St Andrews, Scotland	United Kingdom
Bochum University of Applied Sciences, Department of Geodesy	Bochum	Germany
Carinthian Technical Institute	Villach	Austria
Catholic University of Leuven	Leuven-Heverlee	Belgium
Centro ALGORITMI, Escola de Engenharia, Universidade do Minho	Guimarães	Portugal

Who is AGILE?
<https://agile-gi.eu/>



From **AGILE** to **your community**

3 transferable ideas:

human-centered process, communicative focus, supportive framework of tools and infrastructure.



Human-centered process for reproducibility handles **variety**

Author provides material and makes **fixes**

Collaborate to make things work

Codechecker stops when **confident** enough all parts are provided and then gives review

Communicative focus for reproducibility

Avoids rules & automation playing **catch** with innovation & technology

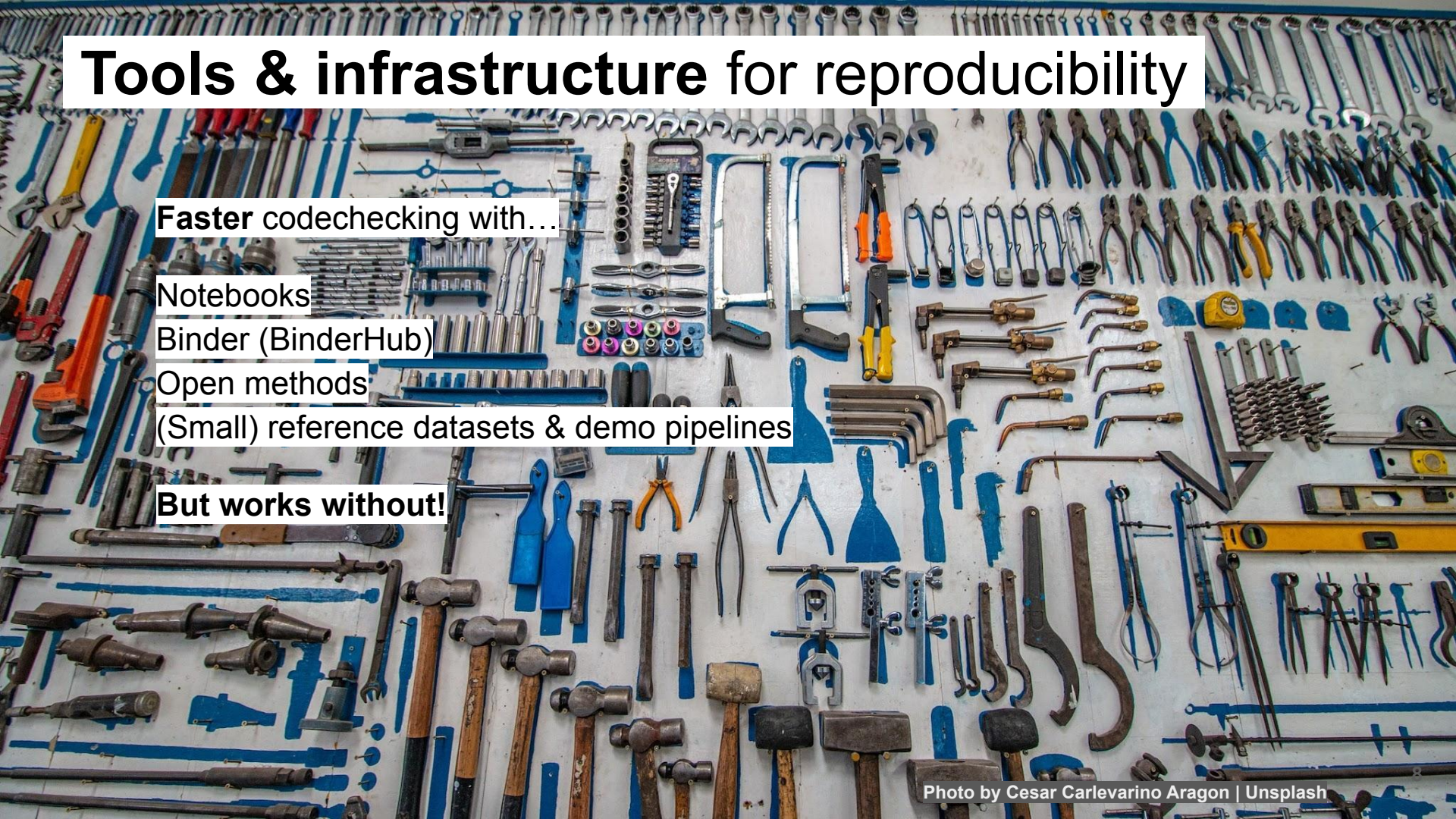
Avoids unification or **limitation** of researcher freedom

Continuous development of shared **practice** over time
& **definition** of “reproducibility” and “how reproducible is enough”

Targets **attitudes** towards sharing, transparency, and openness

Positive learning **experience** for all involved roles

Tools & infrastructure for reproducibility



Faster codechecking with...

Notebooks

Binder (BinderHub)

Open methods

(Small) reference datasets & demo pipelines

But works without!



<https://media.giphy.com/media/mbhseRYedIG5W/giphy.gif>

What are the steps that you could take? Five year plan:

- 1 Initiate **discourse** with a core team in a (small, sub) **community** (e.g., small event/workshop)
- 2 Document **state** of reproducibility & find **leadership** support (metascience paper)
- 3 Establish guidelines (do-ocracy) & **apply** to current work (preprints, post-pub)
- 4 Get **leadership** on board and introduce policy **stepwise**
- 5 **Continue positive** discourse



<https://reproducible-agile.github.io/>

- 2017, '18 & '19: Workshops on reproducibility
- 2019: Reproducible publications at AGILE conferences
(AGILE Initiative supported by AGILE Council)
- 2020: First AGILE reproducibility review
- 2021: Second AGILE reproducibility review
- 2022: Third AGILE reproducibility review - guidelines mandatory
- 2026: Still going, desk reject when no DASA section

Thanks!

Questions?

AGILE Reproducible Paper Guidelines

<https://doi.org/10.17605/OSF.IO/CB7Z8>



(v1)

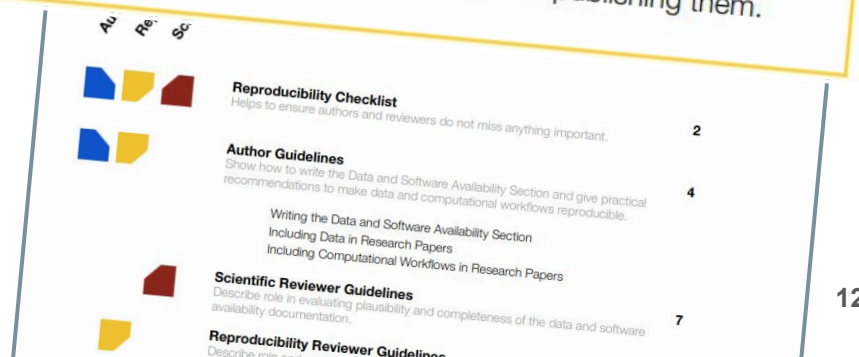
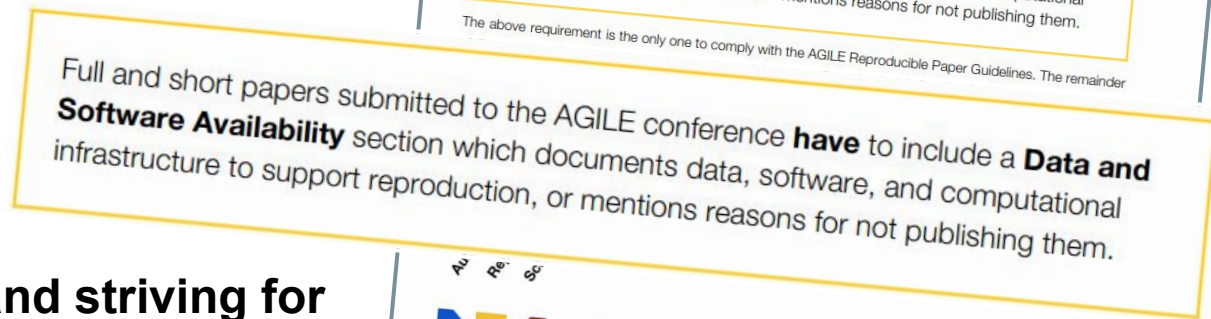
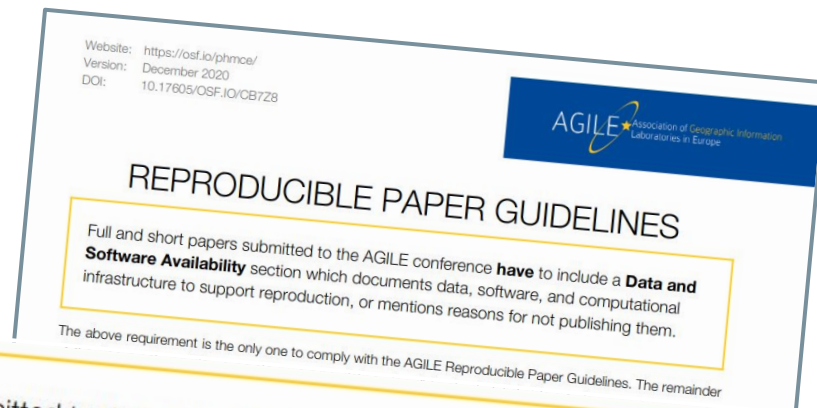
Created by AGILE Initiative in 2019 (see report at <https://osf.io/hupxr/>) and updated in 2020

Transparency over enforcement

Promote, don't exclude

Acknowledge spectrum and striving for ideal

Tailored to GIScience



CODECHECK Certificate 2022-018

10.5281/zenodo.7084333

Raniere Silva

September 27, 2022



Table 1: CODECHECK summary

Title	svaRetro and svaNUMT: Modular packages for annotation of retrotransposed transcripts and nuclear integration of mitochondrial DNA in genome sequencing data
Authors	Ruining Dong, Daniel Cameron, Justin Bodo, Anthony T Papenfuss
Reference	https://doi.org/10.46471/gigabyte.70
Summary	Only visualisation steps performed. All created figures match those in the article.
Repository	https://gitlab.com/cdchck/community-codechecks/2022-svaRetro-svaNUMT.git

Table 2: Summary of output files generated

Files	Comment
figure-2b.pdf	Figure 2(b) of the article
figure-3b.pdf	Figure 3(b) of the article
figure-4.pdf	Figure 4 of the article
figure-5.pdf	Figure 5 of the article
figure-6.pdf	Figure 6 of the article

Summary

The reproduction of the figures, from our R Markdown (.Rmd) files. Figure 3 and 4 are reproduced!

CODECHECKER notes

Data and Code

As a repository was not provided by MANIFEST was created. [Scripts.zip](#) supplemented material in Zenodo (DOI: 10.5281/zenodo.7084333).
\$ make download

Software Installation

The provided .Rmd files requires many bioconductor/bioconductor_docker by running

\$ docker compose up dev

Packages installation instructions are in the README

Running the Script

Figures2-4.Rmd is the main script and history for details. To regenerate the figures as part of the Bioconductor Docker image

Figure 4 uses statistics from `sim_read` `gnomad.Rmd` is the script that renders Figure 4 at the end of the document, see Git history

```
#function from SVEnsemble  
wkdir <- getwd()  
gnomad.bnd.gr <- suppressWarnings(  
  filter(gnomad.bnd.gr, FILTER==  
    hg19_genes,  
    maxgap = 1000,  
    minscore = 0.4  
  )  
)
```

```
gnomad.rt <- rtDetect(  
  filter(gnomad.bnd.gr, FILTER==  
    hg19_genes,  
    maxgap = 1000,  
    minscore = 0.4  
  )  
)
```

takes a couple of hours to execute and
gnomad.insite.pass.rmsk <- gnomad.
find_overlaps(., rmsk.gr, maxgap =
 unique(.) %>%
 as_tibble() %>%
 count(repClass) %>%
 bind_rows(tibble(repClass='non'))

failed with
Error in count(., repClass) : Argument 'repClass' is not a factor
that was resolved by replacing
count(repClass) %>%



References

Dong, Ruining, Daniel Cameron, Justin Bodo, and Anthony T Papenfuss. 2022. "Data and Scripts for the Manuscript of svaRetro and svaNUMT: Modular Packages for Annotating Retrotransposed Transcripts and Nuclear Integration of Mitochondrial DNA in Genome Sequencing Data." Zenodo. <https://doi.org/10.5281/zenodo.7006177>.

Colophon

This document was built with Quarto.

Session Info

```
sessionInfo()
```

R version 4.2.1 (2022-06-23)

Platform: x86_64-pc-linux-gnu (64-bit)

Running under: Ubuntu 22.04.1 LTS

Matrix products: default

BLAS: /usr/lib/x86_64-linux-gnu/libblas.so.3.10.0

LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0

locale:

```
[1] LC_CTYPE=en_GB.UTF-8      LC_NUMERIC=C  
[3] LC_TIME=en_GB.UTF-8      LC_COLLATE=en_GB.UTF-8  
[5] LC_MONETARY=en_GB.UTF-8  LC_MESSAGES=en_GB.UTF-8  
[7] LC_PAPER=en_GB.UTF-8     LC_NAME=C  
[9] LC_ADDRESS=C             LC_TELEPHONE=C  
[11] LC_MEASUREMENT=en_GB.UTF-8 LC_IDENTIFICATION=C
```

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

loaded via a namespace (and not attached):

```
[1] digest_0.6.29  jsonlite_1.8.0  magrittr_2.0.3  evaluate_0.16  
[5] highr_0.9      rlang_1.0.5     stringi_1.7.8   cli_3.4.0  
[9] reticulate_0.14 rmarkdown_2.16  tools_4.2.1     stringr_1.4.1  
[13] xfun_0.33      yaml_2.3.5      fastmap_1.1.0   compiler_4.2.1  
[17] htmltools_0.5.3 knitr_1.40
```

Figures2-4.Rmd's session info:

```
## R version 4.2.1 (2022-06-23)
```

```
## Platform: x86_64-pc-linux-gnu (64-bit)
```

```
## Running under: Ubuntu 20.04.4 LTS
```

```
##
```

```
## Matrix products: default
```

```
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
```

```
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/liblapack.so.3
```

```
##
```

```
## locale:
```

```
## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C  
## [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8  
## [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8  
## [7] LC_PAPER=en_US.UTF-8     LC_NAME=C  
## [9] LC_ADDRESS=C             LC_TELEPHONE=C  
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

codecheck.org.uk/buddy-exchange

Give one, Get one.



Give one, get one CODECHECK! Create your own CODECHECK request and find open CODECHECK issues that need reviewing.

Available Buddy Exchange

CODECHECK Buddies

This list shows all codecheckers who have received buddy exchange requests.

Buddy Exchange Recipients

No buddy exchange requests found.
No codecheckers have received requests.

Statistics

CODECHECK

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team@cdchck.science
github.com/codecheckers
gitlab.com/cdchck
codecheck@fedscience.org
linkedin.com/company/codecheck/
youtube.com/@cdchck

CODECHECK is a process for independent code review of articles.

Need help? Check out the [FAQs](#).

[CODECHECK Register](#) • [Source Code](#)

Register Instance: Production (register)

Next Available Certificate Identifier

2026-005

This is the next available certificate identifier.

How this is calculated:

- We scan all existing issues in the register
- Extract certificate identifiers from issue titles
- Find the next sequential number available

Your issue title will be:

[Author Name] | 2026-005

This will open a new GitHub issue where you can fill in the details.



Create new issue

Blank issue · [Choose a different template](#)

Add a title *

[Author Name] | 2026-005

Add a description

Write

Preview

Type your description here...

CODECHECK Buddy Exchange

Give one, get one CODECHECK! Create your own CODECHECK request and find open CODECHECK issues that need reviewing.

Submit My Request

You are using the testing register. Submissions will go to [test-reg](#), not to production register.

Create your own CODECHECK request to join the buddy exchange.
[Configure my default settings](#) · [View issues in progress](#) · [Update issues](#)

Available Buddy Exchange Issues

#100: [Author Name] 2026-014 by james20 1 month ago Brief Description: Code Repository: Data Repository (if applicable): Published Paper or Preprint: View on GitHub Claim Issue	#107: [Surajad Nussent] 2026-014 by james20 1 month ago Brief Description: Code Repository: Data Repository (if applicable): Published Paper or Preprint: View on GitHub Claim Issue	#56: New CODECHECK 2025-029 by d513m 4 months ago View on GitHub Claim Issue
#46: New CODECHECK 2025-029 by d513m 4 months ago View on GitHub Claim Issue	#42: New CODECHECK 2025-029 by d513m 4 months ago View on GitHub Claim Issue	

CODECHECK Buddies

This list shows all codecheckers who have received buddy exchange requests in the program. Reach out to any of them for potential collaboration!

Buddy Exchange Recipients	Buddy Exchange Contributors
team-codecheck Ratio: 5.0 (received/conducted) ratio	Nussent Last: 8 months ago completed 4
d513m Ratio: 3.1 (received/conducted) PyCharm, PHP ratio	Fosst Last: 8 months ago completed 2
surajad-20 Ratio: 2.0 (received/conducted) ratio	d513m Last: 5 months ago completed 1
Nussent Ratio: 0.0 (received/conducted) github, data analysis, containers, R, Python, JavaScript, Java ratio	Nussent Last: 9 months ago completed 1

Statistics			
6	4	4	Just now
Total Completed	Active Contributors	Recipients Found	Last Updated

CODECHECK Community Workflow

<https://codecheck.org.uk/guide/community-workflow-overview>

For authors

Have a README, find a suitable checklist to use, basic metadata in codecheck.yml, licensing information; acknowledgement in manuscript

For codecheckers (short, extended, in-person events)

Handling repos/files, approaching a workspace/workflow, responsibilities, writing the certificate;

it is a **craft**, so **we mentor** better than we write process descriptions!

For editors (me)



CODECHECK Register

CODECHECK Register

Certificate Number	Year	Topic	Type	Check date	Abstract	Keywords	Conference	Year
2020-001 doi.org/10.17605/OSF.IOX38YR1	2020-03	doi.org/10.17605/OSF.IOX38YR1	Technical Humanitarian Disaster in COVID-19 and Twitter	AGILEDS	Conference	2020-07-13		
2020-002 doi.org/10.17605/OSF.IOX38YR1	2020-04	doi.org/10.17605/OSF.IOX38YR1	Disruptive a multi-agent system for Internet of Things	AGILEDS	Conference	2020-10-07		
2020-003 doi.org/10.17605/OSF.IOX38YR1	2020-05	doi.org/10.17605/OSF.IOX38YR1	The application of Local UCLDs to explore spatial dimensions in archaeological research	J. Archael	Journal	2020-11-19		
2020-004 doi.org/10.17605/OSF.IOX38YR1	2020-06	doi.org/10.17605/OSF.IOX38YR1	Building Change Detection of Address Lines: Spacing and Dense Image Mapping From Clouds using Height and Color Information	AGILEDS	Conference	2021-06-10		
2020-005 doi.org/10.17605/OSF.IOX38YR1	2020-07	doi.org/10.17605/OSF.IOX38YR1	Coastal erosion for coastal line across data a	preprint	Community	2021-04-27		
2020-006 doi.org/10.17605/OSF.IOX38YR1	2020-08	doi.org/10.17605/OSF.IOX38YR1	Relative Spatial	AGILEDS	Conference	2021-07-09		
2020-007 doi.org/10.17605/OSF.IOX38YR1	2020-09	doi.org/10.17605/OSF.IOX38YR1	The Impact of Built Environment on Urban Governance: Along Street Block and Geographically Weighted Models	AGILEDS	Conference	2021-07-09		
2020-008 doi.org/10.17605/OSF.IOX38YR1	2020-10	doi.org/10.17605/OSF.IOX38YR1	Method with a Fuzzy Model of Disturbances	AGILEDS	Conference	2021-07-09		
2020-009 doi.org/10.17605/OSF.IOX38YR1	2020-11	doi.org/10.17605/OSF.IOX38YR1	Open Space Interventions to improve productivity and sustainability of research review preprint	AGILEDS	Conference	2021-07-09		
2020-010 doi.org/10.17605/OSF.IOX38YR1	2020-12	doi.org/10.17605/OSF.IOX38YR1	Sample size estimation for multi-level longitudinal data studies using Bayesian adaptive prior	AGILEDS	Conference	2021-07-09		
2020-011 doi.org/10.17605/OSF.IOX38YR1	2021-01	doi.org/10.17605/OSF.IOX38YR1	Random Data Distribution for Efficient Parallel Point Cloud Processing	AGILEDS	Conference	2021-07-09		
2020-012 doi.org/10.17605/OSF.IOX38YR1	2021-02	doi.org/10.17605/OSF.IOX38YR1	Hydrothermal Freeform Construction in Large-Scale Mass Fabrication	AGILEDS	Conference	2021-07-09		
2020-013 doi.org/10.17605/OSF.IOX38YR1	2021-03	doi.org/10.17605/OSF.IOX38YR1	Knowledge Based Identification of Urban Green Spaces: A Review	AGILEDS	Conference	2021-07-09		
2020-014 doi.org/10.17605/OSF.IOX38YR1	2021-04	doi.org/10.17605/OSF.IOX38YR1	Developing a city specific urban design through a participatory approach	AGILEDS	Conference	2021-07-09		
2020-015 doi.org/10.17605/OSF.IOX38YR1	2021-05	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-016 doi.org/10.17605/OSF.IOX38YR1	2021-06	doi.org/10.17605/OSF.IOX38YR1	Wild Tunnel Load Measurements of a Leading Edge Inflatable Risk Fluid Scale Model	AGILEDS	Conference	2021-07-09		
2020-017 doi.org/10.17605/OSF.IOX38YR1	2021-07	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-018 doi.org/10.17605/OSF.IOX38YR1	2021-08	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-019 doi.org/10.17605/OSF.IOX38YR1	2021-09	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-020 doi.org/10.17605/OSF.IOX38YR1	2021-10	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-021 doi.org/10.17605/OSF.IOX38YR1	2021-11	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-022 doi.org/10.17605/OSF.IOX38YR1	2021-12	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-023 doi.org/10.17605/OSF.IOX38YR1	2022-01	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-024 doi.org/10.17605/OSF.IOX38YR1	2022-02	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-025 doi.org/10.17605/OSF.IOX38YR1	2022-03	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-026 doi.org/10.17605/OSF.IOX38YR1	2022-04	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-027 doi.org/10.17605/OSF.IOX38YR1	2022-05	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-028 doi.org/10.17605/OSF.IOX38YR1	2022-06	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-029 doi.org/10.17605/OSF.IOX38YR1	2022-07	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-030 doi.org/10.17605/OSF.IOX38YR1	2022-08	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-031 doi.org/10.17605/OSF.IOX38YR1	2022-09	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-032 doi.org/10.17605/OSF.IOX38YR1	2022-10	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-033 doi.org/10.17605/OSF.IOX38YR1	2022-11	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-034 doi.org/10.17605/OSF.IOX38YR1	2022-12	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-035 doi.org/10.17605/OSF.IOX38YR1	2023-01	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-036 doi.org/10.17605/OSF.IOX38YR1	2023-02	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-037 doi.org/10.17605/OSF.IOX38YR1	2023-03	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-038 doi.org/10.17605/OSF.IOX38YR1	2023-04	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-039 doi.org/10.17605/OSF.IOX38YR1	2023-05	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-040 doi.org/10.17605/OSF.IOX38YR1	2023-06	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-041 doi.org/10.17605/OSF.IOX38YR1	2023-07	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-042 doi.org/10.17605/OSF.IOX38YR1	2023-08	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-043 doi.org/10.17605/OSF.IOX38YR1	2023-09	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-044 doi.org/10.17605/OSF.IOX38YR1	2023-10	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-045 doi.org/10.17605/OSF.IOX38YR1	2023-11	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-046 doi.org/10.17605/OSF.IOX38YR1	2023-12	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-047 doi.org/10.17605/OSF.IOX38YR1	2024-01	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-048 doi.org/10.17605/OSF.IOX38YR1	2024-02	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
2020-049 doi.org/10.17605/OSF.IOX38YR1	2024-03	doi.org/10.17605/OSF.IOX38YR1	Measuring Landmark Salience in Rural Areas: A Comparative Study of Two Methods	AGILEDS	Conference	2021-07-09		
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